Kubning qirrasini topish

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

---

1. **Kubning hajmi qanday topiladi?**

A) qirrasi × qirrasi
B) qirrasi × 3
C) qirrasi × qirrasi × qirrasi
D) qirrasi × 6

---

2. **Kubning qirrasi 23 cm. Bir yog'ining yuzi qancha?**

A) 12 167 cm²
B) 529 cm²
C) 92 cm²
D) 46 cm²

---

3. $\sqrt{3025}$ **qanday o'qiladi?**

A) «kub ildiz ostida 3025»
B) «3025 ning yarmi»
C) «kvadrat ildiz ostida 3025»
D) «3025 ning kvadrati»

---

4. **Qaysi yozuv «kub ildiz ostida 97 336» deb o'qiladi?**

A) $\sqrt[3]{97\,336}$
B) $\sqrt{97\,336}$
C) $97\,336 \cdot 3$
D) $97\,336 : 3$

---

5. $\sqrt{4900}$ **nechaga teng?**

A) 2450 **B)** 1225 **C)** 700 **D)** 70

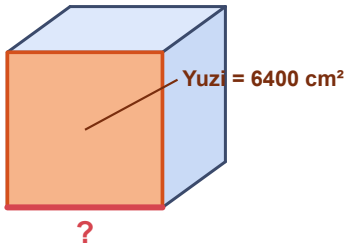
---

6. $\sqrt[3]{64\,000}$ **nechaga teng?**

A) 4000 **B)** 400 **C)** 4 **D)** 40

---

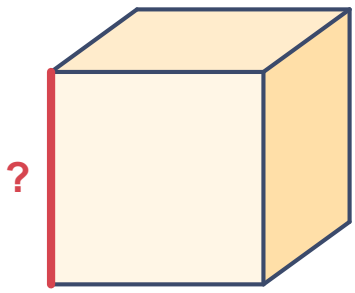
7. **Rasmda kub tasvirlangan, uning bo'yalgan yog'i 6400 cm² ga teng. Kubning bitta qirrasi necha santimetr?**



A) 80 cm
B) 800 cm
C) 3200 cm
D) 1600 cm

---

8. **Rasmdagi kubning hajmi 13 824 m³. Kubning qirrasi qancha?**



A) 2304 m
B) 24 m
C) 4608 m
D) 48 m

---

9. Kubning bir yog'i yuzi ma'lum. Qirrasini topish uchun nima qilinadi?

A) yuzidan kvadrat ildiz olinadi
B) yuzi 2 ga bo'linadi
C) yuzidan kub ildiz olinadi
D) yuzi 4 ga bo'linadi

---

10. Kubning hajmi ma'lum. Qirrasi qanday topiladi?

A) hajm : 3
B) hajm : 6
C) $\sqrt{\text{hajm}}$
D) $\sqrt[3]{\text{hajm}}$

---

11. Qaysi tenglik to'g'ri?

A) $\sqrt{3600} = 600$
B) $\sqrt[3]{512\,000} = 80$
C) $\sqrt{90\,000} = 3000$
D) $\sqrt[3]{512\,000} = 800$

---

12. Nodira kubning hajmi 9261 cm³ ekanini bilib, qirrasini 21 cm deb topdi. Javobni qanday tekshirsa bo'ladi?

A) 21 · 3 = 63
B) 21 · 21 · 21 = 9261
C) 21 + 21 + 21 = 63
D) 21 · 21 = 441

---

13. 27 ta birlik kubchadan bitta kub yasaldi. Kubning bir qirrasida nechta kubcha bor?

A) 3
B) 5
C) 6
D) 9

---

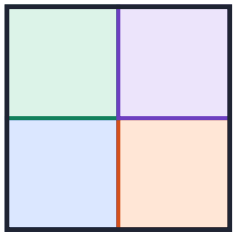
14. Kubning bir yog'i yuzi m² da berilgan. Topilgan qirrasi qaysi birlikda yoziladi?

A) m
B) m³
C) l
D) m²

---

15. Rasmdagi kvadrat 4 ta bir xil kvadratchaga bo'lingan. Katta kvadratning yuzi 1024 cm². Bitta kvadratchaning tomoni qancha?

A) 8 cm
B) 32 cm
C) 16 cm
D) 256 cm



Yuzi = 1024 cm²

---

16. Kvadrat shaklidagi maydonning yuzi 10 000 m². Maydonning tomoni qancha?

A) 1000 m
B) 2500 m
C) 5000 m
D) 100 m

---

17. Kub shaklidagi sandiqning hajmi 729 000 cm³. Sandiqning qirrasi qancha?

A) 900 cm
B) 243 000 cm
C) 9 cm
D) 90 cm

---

18. Kubning qirrasi 20 cm. Uning bir yog'i yuzi va hajmi qancha?

A) 80 cm² va 800 cm³
B) 400 cm² va 1200 cm³
C) 400 cm² va 8000 cm³
D) 60 cm² va 8000 cm³

---

19. Lola kalkulyatorda 2401 sonini yozib, $\sqrt{}$ tugmasini bosdi va 49 chiqdi. Lola kubning qirrasini topayotgan edi. Unga nima berilgan edi?

A) kubning qirrasi 2401 cm

B) kubning hajmi 49 cm³

C) kubning bir yog'i yuzi 2401 cm²

D) kubning hajmi 2401 cm³

---

20. Kubning bir yog'i yuzi 2304 cm². Akbar qirrasini 48 cm deb topdi. Javob to'g'rimi?

A) Ha, chunki 48 · 48 · 48 = 2304

B) Yo'q, chunki 48 · 4 = 192

C) Yo'q, qirrasi 2304 : 4 = 576 cm

D) Ha, chunki 48 · 48 = 2304

Javoblar

Kubning qirrasini topish · 1-test (oson)

№	Javob	Qiymati
1	**C**	qirrasi × qirrasi × qirrasi
2	**B**	529 cm²
3	**C**	«kvadrat ildiz ostida 3025»
4	**A**	$\sqrt[3]{97\,336}$
5	**D**	70
6	**D**	40
7	**A**	80 cm
8	**B**	24 m
9	**A**	yuzidan kvadrat ildiz olinadi
10	**D**	$\sqrt[3]{\text{hajm}}$

№	Javob	Qiymati
11	**B**	$\sqrt[3]{512\,000} = 80$
12	**B**	21 · 21 · 21 = 9261
13	**A**	3
14	**A**	m
15	**C**	16 cm
16	**D**	100 m
17	**D**	90 cm
18	**C**	400 cm² va 8000 cm³
19	**C**	kubning bir yog'i yuzi 2401 cm²
20	**D**	Ha, chunki 48 · 48 = 2304

Qisqa yechimlar

1. Kubning bo'yi, eni va balandligi teng — hammasi qirrasiga teng.
Hajmi = **qirrasi × qirrasi × qirrasi**.
(qirrasi × qirrasi — bu bitta yog'ining yuzi.)

---

2. Kubning yog'i — tomoni 23 cm bo'lgan kvadrat.
Yuzi: 23 · 23 = **529 cm²**.
(12 167 — bu hajm, 92 — 23 · 4, 46 — 23 · 2.)

---

3. $\sqrt{\ }$ — kvadrat ildiz belgisi.
$\sqrt{3025}$ — **«kvadrat ildiz ostida 3025»** deb o'qiladi.

---

4. $\sqrt[3]{\ }$ — kub ildiz belgisi (kichik 3 raqami bilan).
Javob: $\sqrt[3]{97\,336}$. Belgida 3 bo'lmasa — bu kvadrat ildiz.

---

5. Qaysi son o'ziga ko'paytirilsa 4900 chiqadi? 70 · 70 = 4900.
Demak $\sqrt{4900}$ = **70**.
(2450 — 2 ga, 1225 — 4 ga bo'lingan; 700 · 700 = 490 000.)

---

6. 40 · 40 · 40 = 64 000.
Demak $\sqrt[3]{64\,000}$ = **40**.
(4 · 4 · 4 = 64 — nollar unutilgan; 400 · 400 · 400 juda katta.)

---

7. Yog'i — kvadrat, uning tomoni — kubning qirrasi.
$\sqrt{6400}$ = 80, chunki 80 · 80 = 6400. Qirrasi **80 cm**.
(3200 — 2 ga, 1600 — 4 ga bo'lingan.)

---

8. Qirrasi = $\sqrt[3]{\text{hajm}}$.
20 · 20 · 20 = 8000 — kam, 30 · 30 · 30 = 27 000 — ko'p. Sinaymiz: 24 · 24 · 24 = 13 824.
Qirrasi **24 m**. (4608 — 3 ga, 2304 — 6 ga bo'lingan.)

---

9. Kubning yog'i — kvadrat. Kvadratning tomoni yuzining kvadrat ildiziga teng.
Demak **yuzidan kvadrat ildiz olinadi**.

---

10. Kubning qirrasi uzunligi kub ildiz ostidagi hajmga teng: $\sqrt[3]{\text{hajm}}$.
Hajmni 3 ga yoki 6 ga bo'lish qirrani bermaydi.

---

11. 80 · 80 · 80 = 512 000, demak $\sqrt[3]{512\,000} = 80$ to'g'ri.
Qolganlari: $\sqrt{3600}$ = 60, $\sqrt{90\,000}$ = 300.

---

12. Hajmdan qirra topilgan bo'lsa, qirra × qirra × qirra ni hisoblaymiz.
21 · 21 · 21 = 9261 — hajm chiqdi, javob to'g'ri.

---

13. Qirrasidagi kubchalar soni × shu son × shu son = 27 bo'lishi kerak.
3 · 3 · 3 = 27, demak **3** ta.
(9 — 27 ni 3 ga bo'lingan; qirrada 9 ta bo'lsa, kubchalar 729 ta bo'lardi.)

---

14. Qirra — uzunlik. Yuz m² da bo'lsa, qirra **m** da yoziladi.
(m² — yuz, m³ — hajm birligi.)

---

15. Katta kvadrat tomoni: $\sqrt{1024}$ = 32 cm (32 · 32 = 1024).
Har tomonda 2 ta kvadratcha: 32 : 2 = **16 cm**.
(256 — bitta kvadratchaning yuzi, tomoni emas.)

---

16. 100 · 100 = 10 000, demak $\sqrt{10\,000}$ = **100 m**.
(2500 — 4 ga, 5000 — 2 ga bo'lingan; 1000 · 1000 = 1 000 000.)

---

17. 90 · 90 · 90 = 729 000, demak $\sqrt[3]{729\,000}$ = **90 cm**.
(243 000 — 3 ga bo'lingan; 9 — nollar unutilgan: qirrasi 9 cm bo'lsa, hajmi atigi 729 cm³.)

---

18. Bir yog'i: 20 · 20 = 400 cm².
Hajmi: 20 · 20 · 20 = 8000 cm³.
Javob: **400 cm² va 8000 cm³**.

---

19. $\sqrt{\ }$ — kvadrat ildiz. U yog'ning yuzidan qirrani topishda ishlatiladi.
Demak **bir yog'ining yuzi 2401 cm²** berilgan. Hajm berilsa, $\sqrt[3]{\ }$ kerak bo'lardi.

---

20. Yuzdan topilgan qirrani qirra × qirra bilan tekshiramiz: 48 · 48 = 2304.
Demak **javob to'g'ri**. (48 · 48 · 48 = 110 592 — bu hajm bo'lardi.)
